

Supply Chain Database Using SQL

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Table of Contents

Introduction.....	3
Project Objectives	4
Project Overview	4
Database Design.....	5
Suppliers.....	5
Projects	5
Purchase Requisitions	6
Purchase Orders.....	6
Post Order Expediting (POSE).....	7
Freight Companies	7
Freight Estimations	8
Figure 1. Entity Relationship Diagram (ERD) of the Supply Chain Procurement Database.....	9
SQL Implementation.....	10

Introduction

During my internship in the **Supply Chain Department** at **Saipem**, I was introduced to several procurement and logistics processes that support large engineering projects. While I gained practical experience using systems such as **SAP** and worked with procurement documents, supplier information, logistics activities, and post-order expediting, I wanted to strengthen my technical skills by connecting this experience with SQL.

This project was created to simulate a simplified procurement database inspired by real business processes I observed during my internship. The database demonstrates how **SQL** can be used to organize, retrieve, and analyze procurement data through a relational database structure.

The project combines my understanding of supply chain operations with my interest in databases and SQL, allowing me to apply technical skills to a realistic business scenario.

Project Objectives

- Design a relational database using SQL.
- Apply database concepts such as primary keys and foreign keys.
- Simulate procurement and logistics activities using realistic business data.
- Practice writing SQL queries to retrieve meaningful business information.
- Demonstrate how SQL can support decision-making within supply chain operations.

Project Overview

This project models a simplified procurement process inspired by my internship experience in the Supply Chain Department at Saipem. The database represents procurement activities for two Saudi Aramco projects:

- ***Jafurah Gas Development Project***
- ***Berri Offshore Fields***

The database includes supplier information, purchase requisitions, purchase orders, freight companies, freight estimations, and Post Order Expediting (POSE) activities. Although the data used in this project is fictional, it was designed to reflect realistic procurement workflows while demonstrating SQL concepts in a practical business context.

Database Design

The database consists of *seven* relational tables designed to simulate procurement and logistics activities inspired by my internship experience. Each table stores a specific type of information and is connected through primary and foreign keys to maintain data integrity.

Suppliers

Purpose: The **Suppliers** table stores information about companies that provide products or services for procurement activities.

Column	Data Type	Description
SupplierID	INTEGER	Unique identifier for each supplier (Primary Key)
SupplierName	TEXT	Name of the supplier
Country	TEXT	Country where the supplier is located
Category	TEXT	Type of products or services provided

Projects

Purpose: The **Projects** table stores information about the projects associated with procurement activities.

Column	Data Type	Description
ProjectID	INTEGER	Unique identifier for each project (Primary Key)
ProjectName	TEXT	Name of the project
Client	TEXT	Client associated with the project
Location	TEXT	Project location

Purchase Requisitions

Purpose: The **PurchaseRequisitions** table records material requests submitted by different departments before purchase orders are created.

Column	Data Type	Description
PR_ID	INTEGER	Unique purchase requisition number (Primary Key)
ProjectID	INTEGER	References the related project (Foreign Key)
RequestDate	DATE	Date the requisition was created
Department	TEXT	Department requesting the material
MaterialDescription	TEXT	Description of the requested material
RequestedBy	TEXT	Employee or role requesting the material
Status	TEXT	Current status of the requisition

Purchase Orders

Purpose: The **PurchaseOrders** table stores purchase orders issued to suppliers after purchase requisitions are approved.

Column	Data Type	Description
PO_ID	INTEGER	Unique purchase order number (Primary Key)
PR_ID	INTEGER	References the related purchase requisition (Foreign Key)
SupplierID	INTEGER	References the supplier (Foreign Key)
OrderDate	DATE	Date the purchase order was issued
Amount	DECIMAL(10,2)	Total purchase order amount
Currency	TEXT	Currency used for the purchase order
Status	TEXT	Current purchase order status
SABERStatus	TEXT	Status of the SABER certification requirement

Post Order Expediting (POSE)

Purpose: The **PostOrderExpediting** table tracks supplier follow-up activities after purchase orders have been issued.

Column	Data Type	Description
ExpeditingID	INTEGER	Unique identifier for each expediting record (Primary Key)
PO_ID	INTEGER	References the related purchase order (Foreign Key)
FollowUpDate	DATE	Date of supplier follow-up
ExpectedDelivery	DATE	Planned delivery date
ActualDelivery	DATE	Actual delivery date (if completed)
DeliveryStatus	TEXT	Current delivery status
Expeditor	TEXT	Person or team responsible for follow-up
Remarks	TEXT	Additional comments or notes

Freight Companies

Purpose: The **FreightCompanies** table stores information about freight service providers responsible for transporting materials.

Column	Data Type	Description
FreightCompanyID	INTEGER	Unique identifier for each freight company (Primary Key)
CompanyName	TEXT	Name of the freight company

Freight Estimations

Purpose: The **FreightEstimations** table stores estimated transportation information for purchase orders.

Column	Data Type	Description
EstimationID	INTEGER	Unique identifier for each freight estimation (Primary Key)
ProjectID	INTEGER	References the related project (Foreign Key)
PO_ID	INTEGER	References the related purchase order (Foreign Key)
FreightCompanyID	INTEGER	References the freight company (Foreign Key)
TransportMode	TEXT	Transportation method (Sea, Land, etc.)
Origin	TEXT	Shipment origin
Destination	TEXT	Shipment destination
EstimatedCost	DECIMAL(10,2)	Estimated transportation cost
EstimatedTransitDays	INTEGER	Estimated delivery time in days

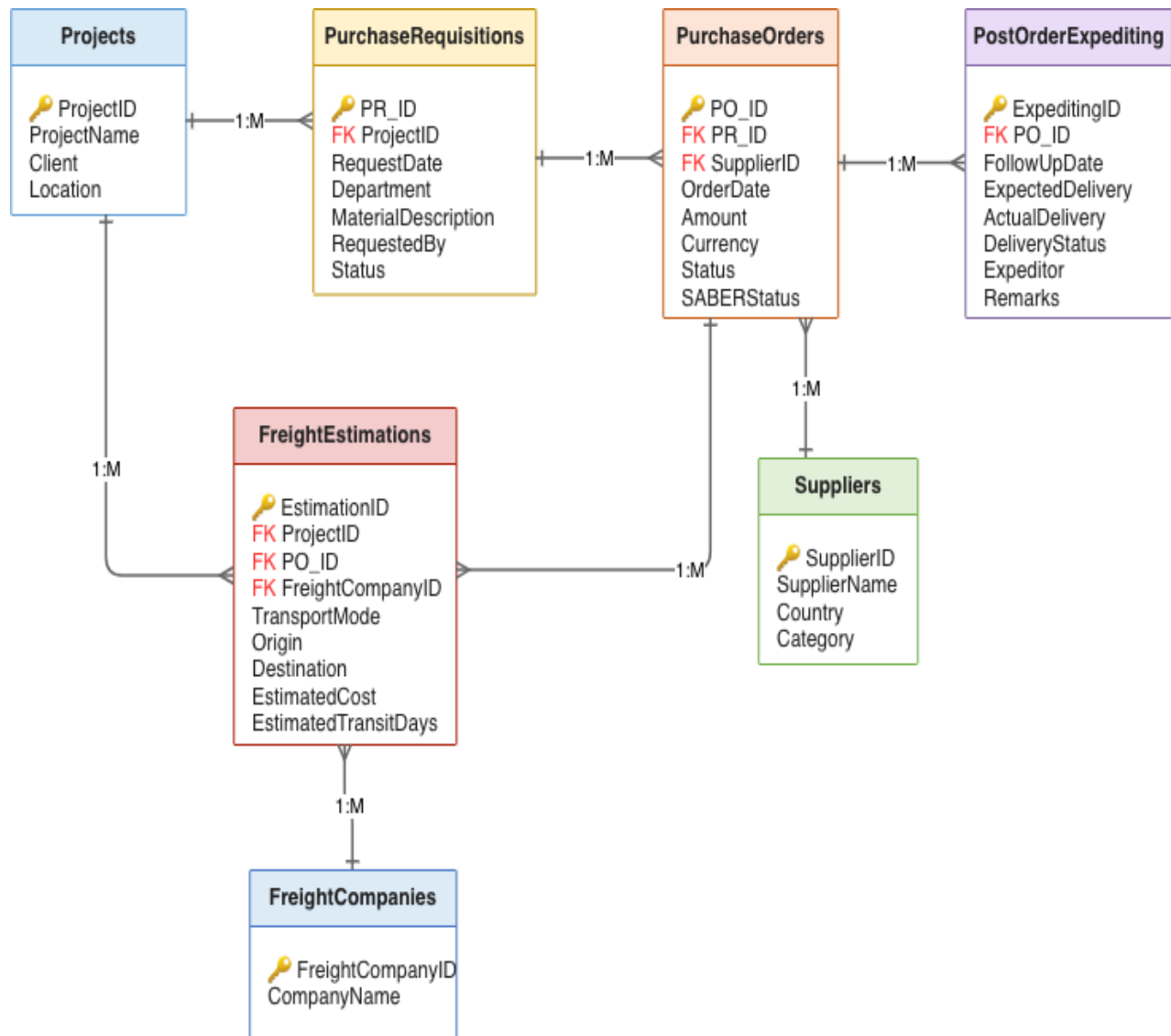


Figure 1. Entity Relationship Diagram (ERD) of the Supply Chain Procurement Database

SQL Implementation

This project was developed using **SQLite Online** to create a relational database that simulates supply chain activities. The implementation involved designing the database schema using ***SQL Data Definition Language (DDL)*** statements and populating the tables with sample data using ***SQL Data Manipulation Language (DML)*** statements.

The database consists of seven related tables connected through *primary* and *foreign* keys to ensure data integrity and accurately represent procurement workflows. The complete SQL scripts used to create and populate the database, and the SQL queries used to retrieve and analyze the data are provided as **separate supporting SQL files**.